Unit: Unit 8: Quadratics **Name:** ________________
Topic: Solving by Factoring and Other Methods **Date:** ___________

Advanced (Carolina Reaper)

Q1. Solve $x^2 + 5x + 6 = 0$ by factoring.
(A) $x = -2, -3$
(B) $x = -1, -6$
(C) $x = 2, 3$
(D) $x = 1, 6$
(E) $x = -2, 3$

Q2. Factor $x^2 - 7x + 12$.
(A) $(x - 3)(x - 4)$
(B) $(x - 2)(x - 5)$
(C) $(x + 3)(x + 4)$
(D) $(x + 2)(x + 5)$
(E) $(x - 1)(x - 6)$

Q3. Solve $x^2 - 4x - 5 = 0$.
(A) $x = -1, 5$
(B) $x = 1, -5$
(C) $x = 2, 3$
(D) $x = -2, 5$
(E) $x = 5, -1$

Q4. Factor $x^2 + 2x - 15$.
(A) $(x + 5)(x - 3)$
(B) $(x + 2)(x - 5)$
(C) $(x - 3)(x + 5)$
(D) $(x + 3)(x - 5)$
(E) $(x + 1)(x - 3)$

Q5. Solve $x^2 - 9x + 20 = 0$ by factoring.
(A) $x = 4, 5$
(B) $x = 5, 4$
(C) $x = 4, 9$
(D) $x = 2, 10$
(E) $x = 9, 4$

Q6. Solve $x^2 + x - 6 = 0$.
(A) $x = -3, 2$
(B) $x = 3, -2$
(C) $x = -1, 6$
(D) $x = 1, -6$
(E) $x = 2, 3$

Q7. Factor $x^2 - 2x - 15$.
(A) $(x - 5)(x + 3)$
(B) $(x - 3)(x + 5)$
(C) $(x + 1)(x - 3)$
(D) $(x + 2)(x - 5)$
(E) $(x + 5)(x - 3)$

Q8. Solve $x^2 - x - 12 = 0$.
(A) $x = -3, 4$
(B) $x = -4, 3$
(C) $x = 3, 4$
(D) $x = 4, -3$
(E) $x = 1, -12$

Open Ended Questions (FRQ)

Q9. Solve $x^2 + 4x + 4 = 0$ and verify the solution.

Q10. Factor $x^2 - 3x - 10$ and solve the equation $x^2 - 3x - 10 = 0$.

Chef's Tips

- Provide clear and concise explanations for each solution.
- Make sure students show all steps in the factoring process.
- Emphasize the importance of verifying solutions.